

You have learnt that all matter on earth consists of elements. Elements are simplest kind of matter which cannot be further split up into simpler substances by chemical reactions. The smallest particle of an element is called an atom. Let us explore what are atoms, what do they contain and how do they behave?

In this Chapter you will learn about:

- Structure of an atom (Protons, neutrons and electrons)
- Atomic Number and Mass Number
- Distribution of Electrons in shells (K,L,M only) using $2n^2$ Formula (1-18 Elements)
- Valency and Ions
- Isotopes and their uses
- Chemical Formulae
- Law of constant composition

All the students will be able to:

- ✓ Describe the Structure of an atom.
- ✓ Differentiate between atomic number and mass numbers.
- ✓ Draw diagrams of the atomic structure of the first eighteen elements in the periodic table.
- ✓ Define valency.
- ✓ Explain formation of ions.
- ✓ Differentiate between cations and anions.
- ✓ Describe isotopes and their uses in medicine and agriculture.
- ✓ Identify the types and number of elements present in simple molecules and compounds.
- ✓ Make chemical formulae from list of anions and cations.
- ✓ State the law of constant composition and give examples.

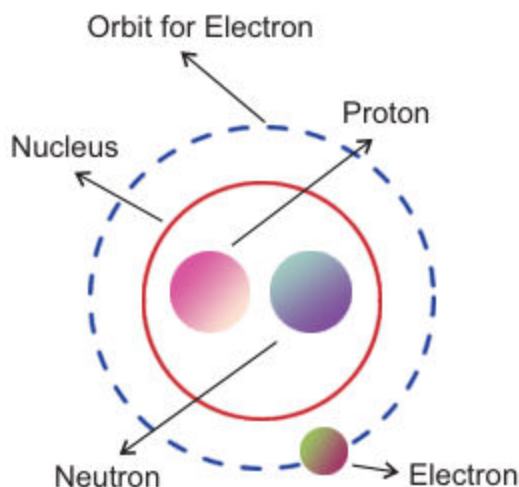


Fig 6.1
Structure of an Atom



Fig 6.2 Everything is made up of different kinds of atoms

STRUCTURE OF AN ATOM

✓ Describe the Structure of an atom.

An atom is an extremely small particle of matter that cannot be subdivided by ordinary means. The atoms of different elements have different sizes. However, each atom is made up of smaller particles called “subatomic particles” i.e. protons, neutrons and electrons. Atoms have equal number of protons and electrons e.g. an atom of hydrogen has one proton and one electron, an atom of nitrogen has seven protons and seven electrons.

Nucleus:

All atoms have a small dense positively charged nucleus in their centers. All the mass or weight of an atom is concentrated in the nucleus due to the presence of neutrons and protons which are about 1836 times heavier than electrons. The nucleus is very much smaller than the atom as a whole.

Proton:

Proton has positive charge and its mass is one atomic mass unit (1amu). The number of protons in an atom is always equal to the number of electrons in it.

Neutron:

Neutron has no charge. Protons and neutrons have almost same mass 1amu. They are collectively known as nucleons.

Electron:

The electrons are arranged in shells around the nucleus. They revolve or whirl in these fixed shells or orbits around the nucleus. The electrons have a negative electric charge. An atom has the same number of electrons and protons.

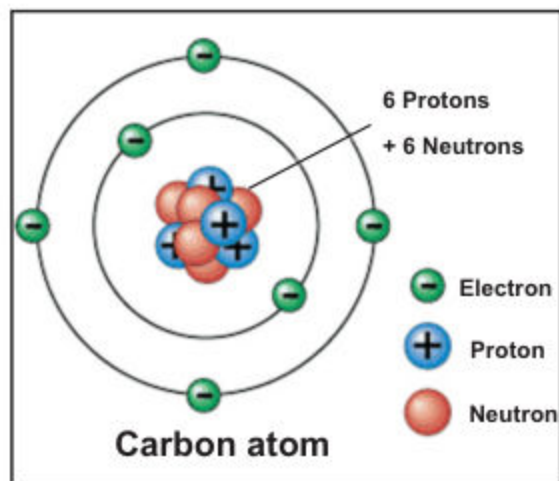


Fig 6.3 Structure of carbon atom

ATOMIC NUMBER AND MASS NUMBER

✓ Differentiate between atomic number and mass number.

Atomic Number:

The specific number of protons present in the nucleus of every atom is called its atomic number. It is represented by the symbol 'Z'. Carbon atom has 6 protons inside the nucleus see Fig 6.3. Therefore, its atomic number is 6. The number of protons are also equal to the number of electrons present in an atom.

Mass Number:

Mass number is the sum of protons and neutrons in the nucleus of an atom. It is represented by "A". Again observe figure 6.3, where the carbon atom has 6 protons and 6 neutrons in its nucleus so, the total number of protons and neutrons become 12. Therefore, the mass number or the nucleon number of carbon is 12.

$$A = Z + N$$

Number of protons (Z) = 6

Number of neutrons (N) = 6

$$\text{Mass number of Carbon (A)} = 6 + 6 = 12$$

The difference between the mass number 'A' and the atomic number 'Z' of an atom gives the number of neutrons 'N'. Thus,

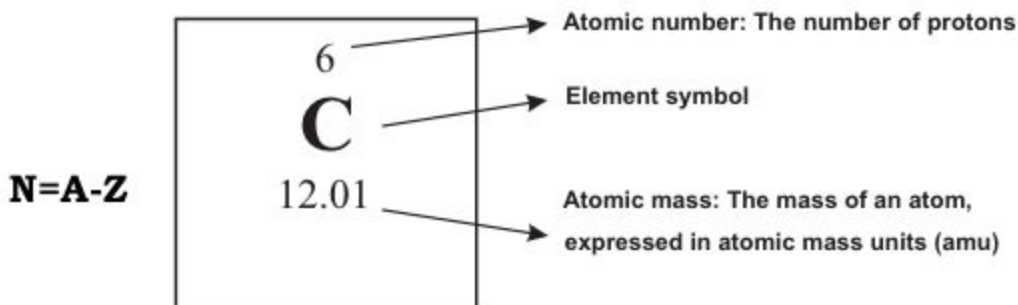


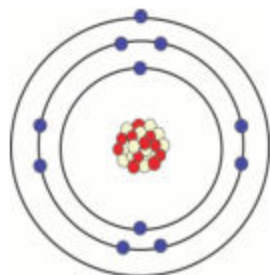
Figure 6.4 Representation of an element Carbon and its atomic and mass numbers

Activity 6.1: Answer the following questions by observing the following figure and using the periodic table:

Number of Electrons = 11

Number of Neutron = 12

Number of Proton = ?



The given atomic structure represents which element? _____

Write its atomic number and mass number: $Z =$ _____; $A =$ _____

Write the number of protons _____.

THE PERIODIC TABLE

- ✓ Draw diagrams of the atomic structure of the first eighteen elements in the periodic table.

DO YOU KNOW?

We can observe surface of an object at atomic level with special type of microscope called, the Scanning Tunneling Microscope (STM) developed in 1981.

Periodic table is a chart of all the known elements in which elements are categorized into groups of similar chemical properties in the order of increasing atomic numbers (See figure 6.5). There are rows of different colours and they go from left to right. Row is considered to be a period. All the periods in a row has same number of atomic shells. The periodic table has columns from top to bottom called groups. The elements in a group have the same number of electrons in their outer most shell.

Part of the periodic table							
1 H							4 He
7 Li	9 Be	11 B	12 C	14 N	16 O	19 F	20 Ne
23 Na	24 Mg	27 Al	28 Si	31 P	32 S	35.5 Cl	40 Ar

Key atomic mass — 4
atomic number — 2
He

Figure 6.5 Part of periodic table

Activity 6.2: Draw diagrams of atomic structure of first 18 elements in the periodic table

What I need:

- Periodic table
- Exercise Book
- Coloured pencils/ markers
- Compass for drawing circles

What to do:

- Observe the Fig 6.5 part of periodic table.
- Draw diagrams of all 18 elements in your note book with the help of atomic number and mass number.
- Show it to the teacher and share with the whole class.

VALENCY

✓ Define valency.

Valency is the power of combining of an atom with other atoms to form molecules or compounds .

An atom may need to gain or lose an electron from its outermost or valence shell, which has to complete its outermost shell, so that it will be like noble gases. This tendency to lose or gain an electron is also called valency.

Sodium has a valency of 1 because it has tendency to lose one electron from its valence shell (figure 6.6(a)).

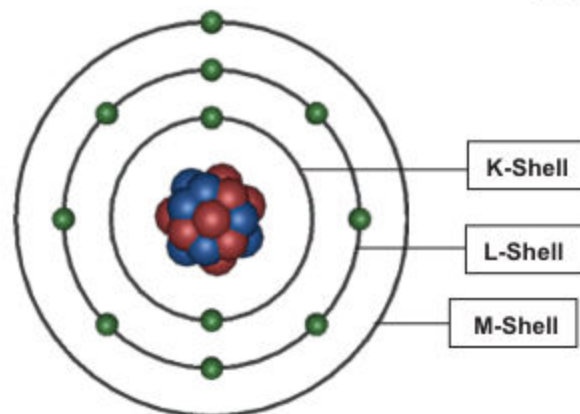


Figure 6.6 a) Atomic structure of sodium

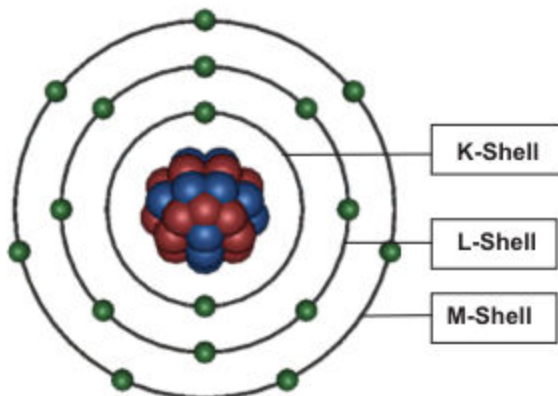


Figure 6.6 b) Atomic structure of chlorine

Chlorine has a valency 1 because it has tendency to gain one electron. If it gains an electron its valence shell becomes complete (figure 6.6(b)).

Distribution of electrons in shells using $2n^2$ formula:

Electrons revolve around the nucleus in definite paths called orbits or shells. Each shell can contain only a fixed number of electrons. The formula ' $2n^2$ ' is used to calculate the maximum number of electrons that an orbit can possess where ' n ' indicates the position of the orbit from nucleus.

Number of orbit(n)	Names of orbits	$2n^2$	Maximum number of electrons
1	K	2×1^2	2
2	L	2×2^2	8
3	M	2×3^2	18

The first orbit, which is closest to nucleus is called K shell. The second shell is called L shell, likewise the third shell is called M shell.

IONS

- ✓ Explain formation of ions.
- ✓ Differentiate between cations and anions.

An atom is neutral but when an atom or molecule that carries net electric charge due to gain or loss of an electron is called an ion. Atoms which lose electrons will become positively charged ion and are known as cations. (figure 6.7 (a)). The size of a cation is smaller than that of its parent neutral atom.

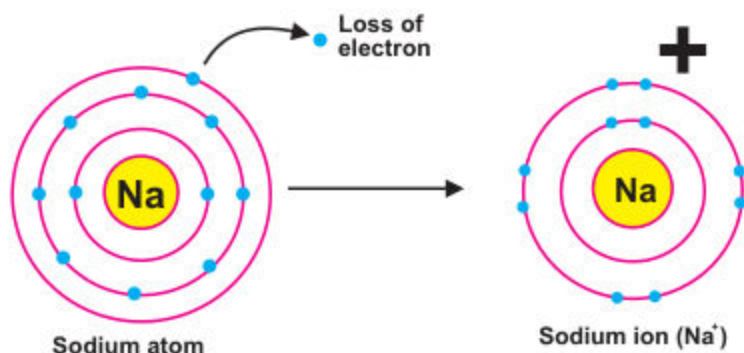


Figure 6.7. (a) Formation of sodium ion

Atoms which gain electron will become negatively charged and form negative ion which is called anion. As shown in Figure 6.7 (b). The size of an anion is greater than that of its parent neutral atom.

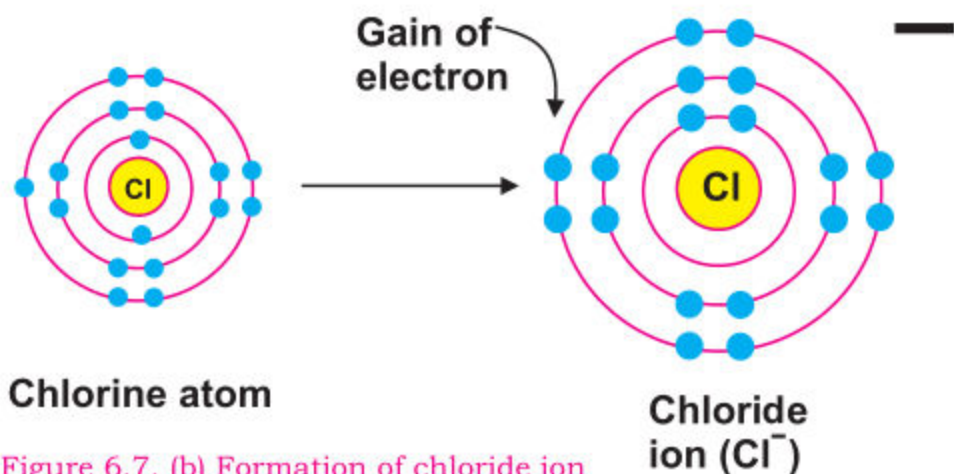


Figure 6.7. (b) Formation of chloride ion

✓ Make chemical formulae from list of anions and cations.

The table 6.1 shows some common anions and cations.

Charge of ion	Name of ion	Symbol of ion	Number of electrons lost or gained by atom
+1	Hydrogen	H^+	Lost one electron
	Lithium	Li^+	
	Sodium	Na^+	
+2	Beryllium	Be^{++}	Lost two electrons
	Magnesium	Mg^{++}	
+3	Boron	B^{+++}	Lost three electrons
	Aluminum	Al^{+++}	
-1	Chloride	Cl^-	Gained one electron
	Fluoride	F^-	
-2	Oxide	O^{--}	Gained two electrons
	Sulphide	S^{--}	

Activity 6.3: Complete the following table

Name of Compound	Chemical Formula	Elements	No. of Atoms of each Element
Carbon mono oxide	CO	Carbon, Oxygen	1 Carbon, 1 Oxygen
Ammonia	NH ₃		
Water	H ₂ O		
Sodium chloride	NaCl		
Calcium chloride	CaCl ₂		

Activity 6.4:

Write the names and chemical formulae of at least 10 molecules with the help of list of cations and anions given in Table 6.1

- | | |
|----------|-----------|
| 1. _____ | 6. _____ |
| 2. _____ | 7. _____ |
| 3. _____ | 8. _____ |
| 4. _____ | 9. _____ |
| 5. _____ | 10. _____ |

ISOTOPES

✓ Describe isotopes and their uses in medicine and agriculture.

The atoms of same element which have the same atomic number but different mass numbers are known as “isotopes”.

Isotopes are represented by following symbol:



X = element symbol

Z = atomic number (number of protons)

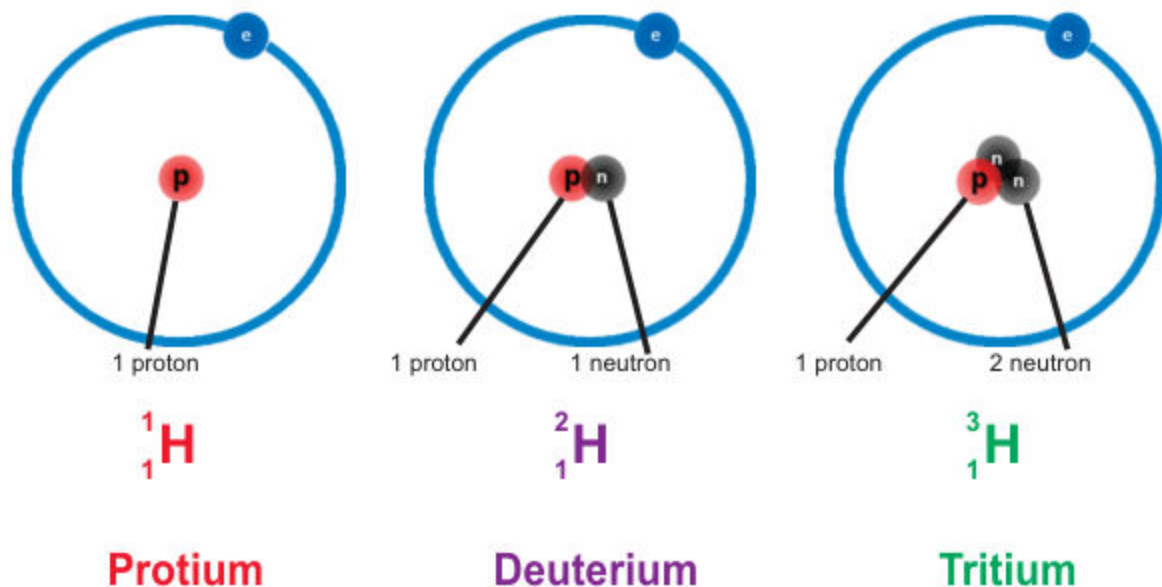
A = mass number (number of protons + number of neutrons)



Figure: 6.8 (a) Isotopes of hydrogen

Teacher Note: Teacher should make groups or pair of students according to class size and facilitate them to work collaboratively by sharing ideas and actively working together.

Thus, ordinary hydrogen or protium is written as ${}^1_1\text{H}$, deuterium is isotope of hydrogen written as ${}^2_1\text{H}$ and tritium as ${}^3_1\text{H}$.



6.8 (b). Isotopes of Hydrogen- Protium, deuterium and tritium

The atomic structures of carbon isotopes are shown in the figure 6.8(c). Carbon atoms exist naturally with 6, 7 or 8 neutrons. Since each atom of carbon has 6 protons, its isotopes have atomic masses 12, 13 and 14. These isotopes are called carbon-12, carbon-13 and carbon-14, respectively. Alternatively, they may be written as ${}^{12}_6\text{C}$, ${}^{13}_6\text{C}$ and ${}^{14}_6\text{C}$.

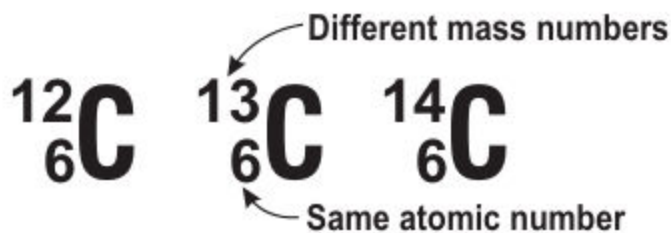


Fig 6.8 (c) Isotopes of carbon

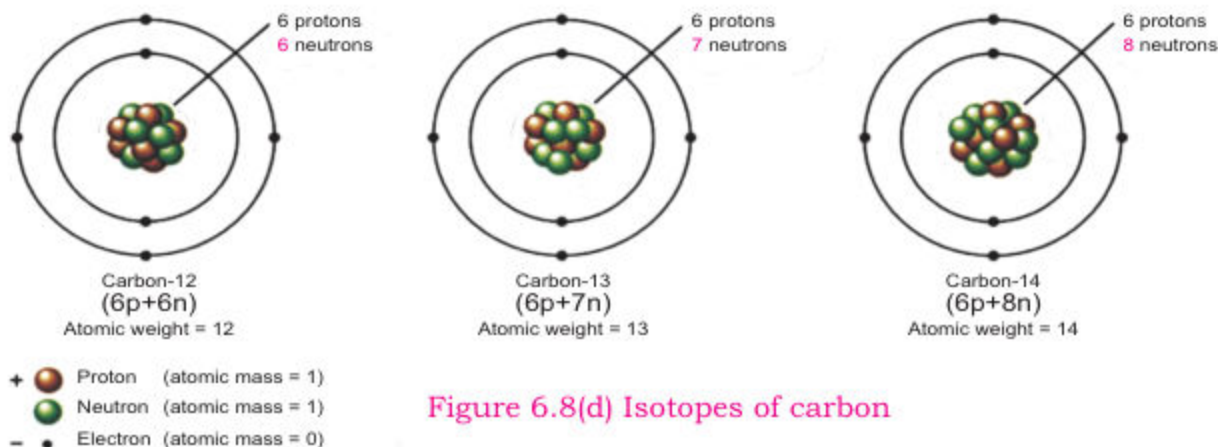


Figure 6.8(d) Isotopes of carbon

Uses of isotopes:

Uses in Medicine:

Sr. No	Isotopes (Symbols)	Uses
1.	Copper – 64 (^{64}Cu)	Used for diagnosis and treatment of various diseases, for example in Wilson's disease, which is a hereditary disorder. This isotope helps in detection of copper retention in the body. Copper isotope is used to kill cancer cells by a particular type of treatment, known as radiotherapy.
2.	Gallium – 67	Used for detection of prolonged infection and cancer cells .
3.	Cobalt – 60	Used for radiotherapy to stop the growth of cancer cells.
4.	Iron – 59	Used in studies of iron metabolism and proper functioning of red blood cells (RBCs).
5.	Iodine – 123	Used for diagnosis of function of thyroid gland.

Uses in Agriculture:

Sr. No.	Isotopes	Uses
1.	Nitrogen – 15	Used to identify efficiency rates of organic and inorganic fertilizers.
2.	Phosphorous – 32	Used to study the uptake of phosphorous containing fertilizers.
3.	Carbon – 14	Used to investigate carbon uptake and biochemical pathways in plants. Carbon dating is also used for the estimation of the age of fossils.
4.	Lead – 210	Used to determine the accumulation rates of pollutants in layers of soil.
5.	Iron Copper Cobalt	Used to determine quantitatively the micro-nutrient uptake by plants

MOLECULE

- ✓ Identify the types and number of elements present in simple molecules and compounds.

A molecule is formed when two or more same or different atoms join together chemically. Some molecules consist of two atoms of the same element (O_2, Cl_2), while other molecules like water (H_2O), methane (CH_4) etc consist of different atoms. Simple molecules of oxygen (O_2) and chlorine (Cl_2) are shown in figure 6.9.

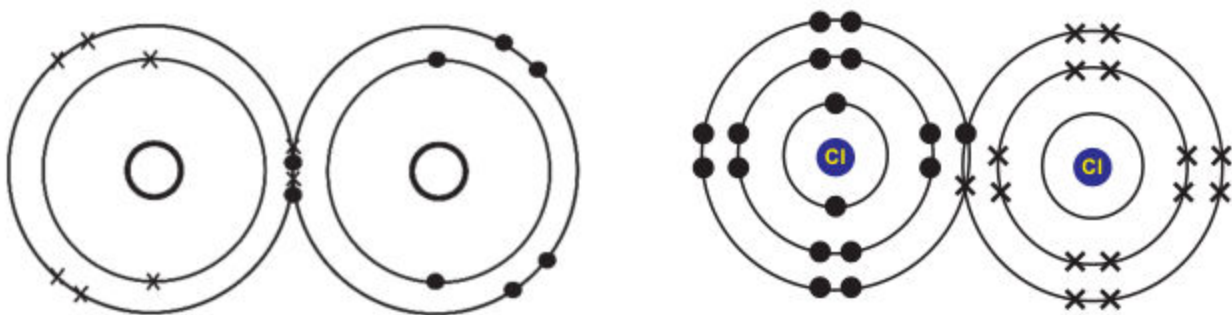


Figure 6.9 Molecules of oxygen and chlorine

Compound:

Compound is a molecule that contains atoms of at least two different elements.

Water (H_2O), carbon dioxide (CO_2) and methane (CH_4) are compounds because each is made from more than one element. Water (H_2O) is composed of two atoms of hydrogen and one atom of oxygen.

The simple compound of hydrochloric acid (HCl) and sodium chloride ($NaCl$) are shown in figure 6.10.



Figure 6.10 Hydrochloric acid (HCl) and sodium chloride $NaCl$ molecules/compounds

CHEMICAL FORMULA

- ✓ Make chemical formulae from list of anions and cations.

The symbolic representation of molecule or compound is called chemical formula. The formula shows the constituent elements and their exact number of atoms present in a molecule or a compound.

The chemical formula of water is explained below:



Two hydrogen atoms are present in one molecule of water.

Only one oxygen atom is present in one water molecule.

Chemical formula of water

Similarly, when sodium and chloride ions are combined in the ratio of 1:1, we get common salt which has the chemical formula NaCl.

LAW OF CONSTANT COMPOSITION

- ✓ State the law of constant composition and give examples.

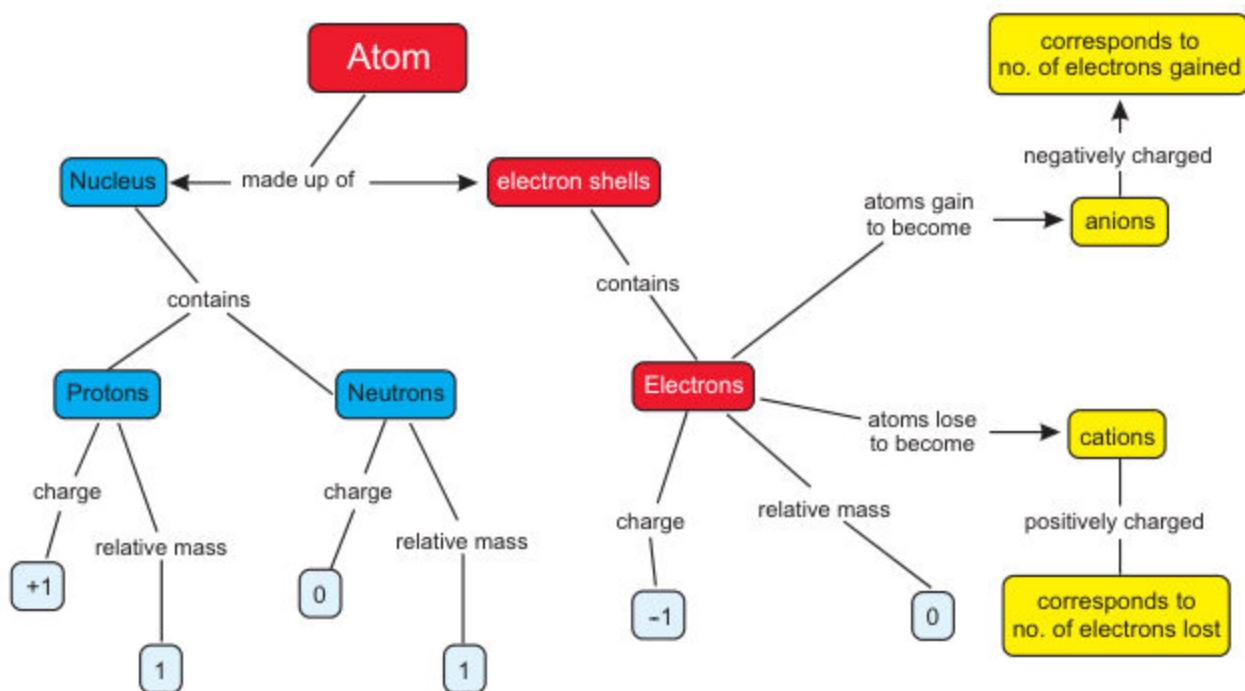
"A chemical compound always contains the same elements combined together in the same proportion or ratio." For example, clean water obtained from different sources such as river, well, spring, sea, etc., always made up of 2 hydrogen atoms and one oxygen atom in the ratio of 2:1. Similarly, carbon dioxide (CO_2) can be obtained by different methods such as: burning of carbon, heating of lime stone, reaction of HCl with marble pieces.

However, each sample of CO_2 contains carbon and oxygen in the ratio of 1:2

Similarly, in any compound, all samples of that compound will be made up of the same elements in the same proportion or ratio with each other. When elements combine, they combine in the ratio of small whole number. For example any water molecule is always made up of two Hydrogen atoms and one Oxygen atom in a ratio 2:1.

The formula of calcium chloride is CaCl_2 ; the ratio of Ca to Cl is 1:2.
The formula of Magnesium oxide is MgO ; the ratio of Mg to O is 1:1.

Summary



Review Exercises

1. Write short answers of the following questions:
 - i) Which particles are found within the nucleus of an atom? What electrical charges do they carry?
 - ii) Name particles which revolve around the nucleus of an atom?
 - iii) Why atom is neutral?
 - iv) Draw the atomic structures of the following elements:
(a) Magnesium (b) Silicon (c) Sulphur (d) Calcium (e) Aluminium
 - v) Differentiate between cations and anions.

2. Complete the table below.

Sr. No.	Atom	Ion	No. of protons in ion	No. of electrons in ion
1.	Chlorine			
2.	Sodium			
3.	Potassium			

3. How many electrons do the following elements have in their valence shells? Write down their valencies.

(a) Nitrogen (b) Helium (c) Lithium.

4. Which element has one electron in its outermost shell?

5. Write down the number of carbon and hydrogen atoms in methane?

6. Write down the chemical formula of ammonia molecule, which contains one nitrogen and three hydrogen atoms.

7. Complete the table:

S. No	Compounds	Chemical formula	Elements	Ratios between atoms
1	Water	H ₂ O		
2	Aluminium oxide	Al ₂ O ₃		
3	Calcium oxide	CaO		
4	Hydrochloric acid	HCl		
5	Calcium fluoride	CaF ₂		

8. Choose the correct answer from the given choices.

a) The entire mass of an atom is concentrated in the center of an atom called _____.

Nucleus, Orbit, Neutron.

b) The mass of an atom is determined by _____.

Neutron + Electron, Electron + Proton, Neutron + Proton.

- c) Atoms of same element having different mass number are called _____.

Anions, Cations, Isotopes

- d) Chemical formula shows _____.

Number of atoms in a molecule, Number of atoms of each element, Number of atoms in a compound

- e) An atom has atomic number 16 and mass number 32, the number of protons is _____. 16, 32, 18

9. Fill in the following chart.

Atomic number	Mass number	Number of protons	Number of neutrons	Number of electrons	Symbol of element	Isotope
6	12					
	24	12				
	35			17		
13			14			

PROJECT

MODEL MAKING

What I need:

Beads/small size balls of two different colors to represent different elements.

Wires to make orbits

What to do:

Show atomic structures of calcium, magnesium, sodium and chlorine atoms by preparing models.